

ECA0303 SIGNALS AND SYSTEM

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Experiment 4.2

Causal/Non-Causal System

Program :

% Test Case: Causal and Non-Causal Systems Analysis

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Define Parameters
```

```
N = 20;           % Number of samples
```

```
n = 0:N-1;       % Discrete time index vector
```

```
x_n = ones(1, N); % Input sequence: Unit step sequence
```

```
%% Initialize Output Sequences
```

```
y_causal = zeros(1, N); % Output sequence for causal system
```

```
y_noncausal = zeros(1, N); % Output sequence for non-causal system
```

```
%% Causal System:  $y[n] = x[n] + 0.5 \cdot x[n-1]$ 
```

```
for k = 1:N
```

```
    if k > 1
```

```
        y_causal(k) = x_n(k) + 0.5 * x_n(k-1);
```

```
    else
```

```
        y_causal(k) = x_n(k); % No  $x[n-1]$  for the first sample
```

```
    end
```

```
end
```

```

%% Non-Causal System:  $y[n] = 0.5x[n] + 0.5x[n+1]$ 
for k = 1:N
    if k < N
        y_noncausal(k) = 0.5 * x_n(k) + 0.5 * x_n(k+1);
    else
        y_noncausal(k) = 0.5 * x_n(k); % No x[n+1] for the last sample
    end
end
end

```

```

%% Plot the Input Sequence

```

```

figure;

subplot(3,1,1);
stem(n, x_n, 'filled');
grid on;
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Input Sequence: Unit Step Sequence');

```

```

%% Plot the Output of the Causal System

```

```

subplot(3,1,2);
stem(n, y_causal, 'filled');
grid on;
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Output Sequence: Causal System');

```

```

%% Plot the Output of the Non-Causal System

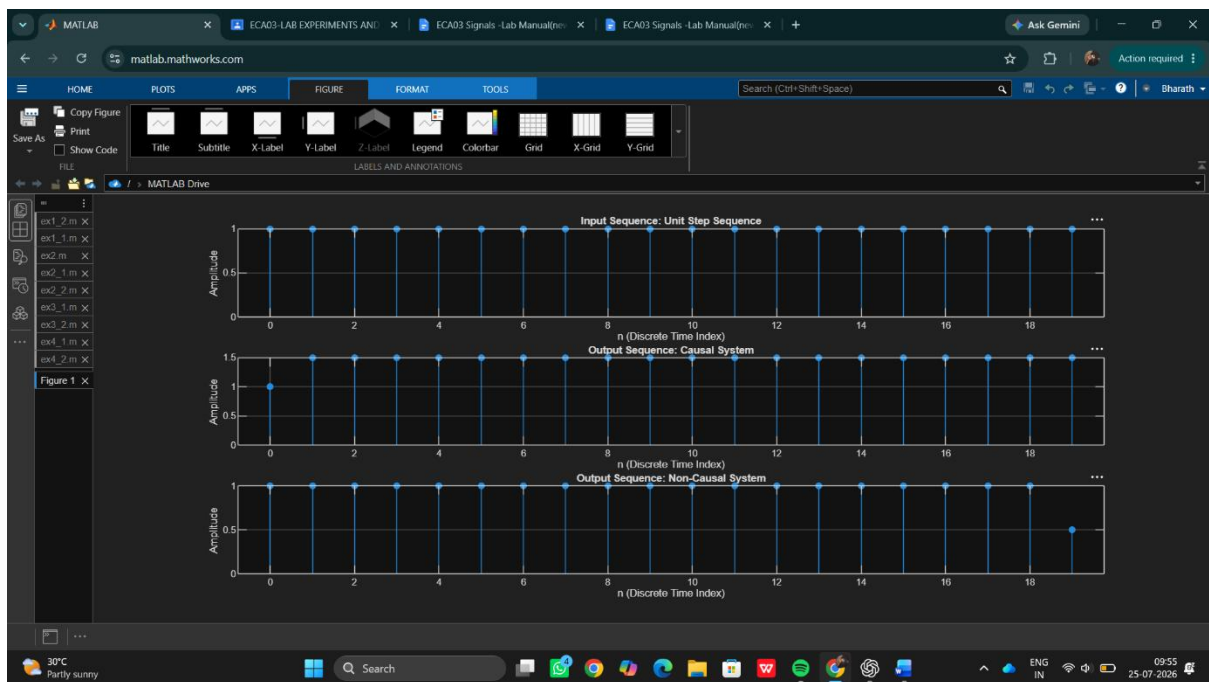
```

```

subplot(3,1,3);
stem(n, y_noncausal, 'filled');
grid on;
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Output Sequence: Non-Causal System');

```

Output waveform :



Result:

- **Causal System:** The causal system only depends on the current and past input values. The output does not require future input values, making it physically realizable in real-time systems.
- **Non-Causal System:** The non-causal system depends on future input values, which is not physically realizable in real-time systems but useful for certain theoretical analyses.

This test case helps to illustrate the difference between causal and non-causal systems, demonstrating how their characteristics affect the response to a given input sequence.